

# Current and Future Estimates of Marginal Emission Factors for Indian Power Generation

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Cite This: *Environ. Sci. Technol.* 2022, 56, 9237–9250



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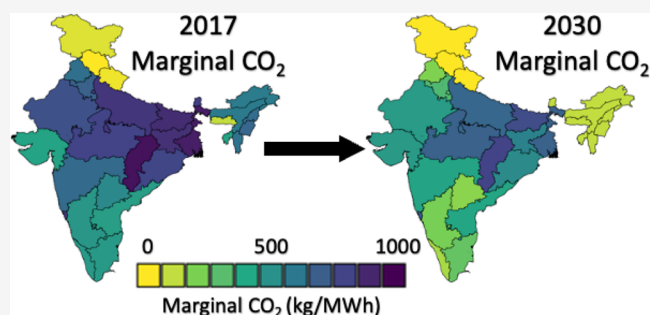
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**ABSTRACT:** Emission factors from Indian electricity remain poorly characterized, despite known spatial and temporal variability. Limited publicly available emissions and generation data at sufficient detail make it difficult to understand the consequences of emissions to climate change and air pollution, potentially missing cost-effective policy designs for the world's third largest power grid. We use reduced-form and full-form power dispatch models to quantify current (2017–2018) and future (2030–2031) marginal CO<sub>2</sub>, SO<sub>2</sub>, NO<sub>x</sub>, and PM<sub>2.5</sub> emission factors from Indian power generation. These marginal emissions represent emissions changes due to small changes in demand. For 2017–2018, spatial variability in marginal CO<sub>2</sub> emission factors range 3 orders of magnitude across India's states. There is limited seasonal and intraday variability with coal generation likely to meet changes in demand more than half the time in more than half of the states. Assuming the Government of India approximate 2030 targets, the median marginal CO<sub>2</sub> emission factor across states decreases by approximately a factor of 2, but emission factors still span 3 orders of magnitude across states. Under 2030–2031 assumptions there is greater seasonal and intraday variability by up to factors of two and four, respectively. Estimates provide emission factors to evaluate interventions such as electric vehicles, increased air conditioning, and energy efficiency.

**KEYWORDS:** India, emissions, marginal, electricity, air pollution, climate



## INTRODUCTION

As the world's third-largest emitter of greenhouse gases (GHG)<sup>1–3</sup> and a large emitter of local criteria air pollutants that lead to premature mortality, India's current emissions are consequential to global efforts to mitigate climate change and improve air quality. Current per capita emissions remain a fraction of the world average but growing.<sup>4</sup> Forty percent of the country's GHG emissions come from its coal-heavy power sector,<sup>5</sup> the world's third-largest power sector in terms of electricity generation across all nations.<sup>3</sup> Likewise, uncontrolled sulfur dioxide (SO<sub>2</sub>) and nitrogen oxides (NO<sub>x</sub>) emissions from Indian power generation contribute to 7–21% of up to 1.1 million premature deaths in India associated with ambient PM<sub>2.5</sub> pollution.<sup>6–13</sup>

Future scenarios of India's power sector with current stated policies show the power sector would remain a consistent, significant source of GHG emissions and air pollution.<sup>4,14,15</sup> Expected increases in electricity demand from economic growth drive these future emissions. Point source emitters, such as the power sector, are generally policymakers' focus for emissions control policies. The Government of India has a 2030 target to increase renewable and other zero-emission capacity to 50% of the country's electricity mix.<sup>16,17</sup>

Emissions factors from the Indian power grid remain poorly characterized, which may make wise and cost-effective decision making more difficult. These emission factors provide policymakers and analysts a basis to evaluate decarbonization efforts and other interventions. A number of studies have characterized GHG and air pollutant emissions nationally or by individual plants in India as well as their impacts on ambient air quality.<sup>6,11,13,14,18–23</sup> However, none have characterized emission factors spatially and temporally based on electricity consumption, i.e., the emissions associated with power used in different areas of the country.

Complicating efforts are the lack of widely available continuous emissions monitoring system (CEMS) data from Indian power plants,<sup>24</sup> the decentralized, federal system of power scheduling and dispatch in India, and high transmission and distribution losses that vary by state in India. Individual

**Received:** November 3, 2021

**Revised:** June 6, 2022

**Accepted:** June 8, 2022

**Published:** June 24, 2022



states are balancing areas that contract with and schedule power (“self-scheduling”) from various portfolios of plants owned by the central government, by state governments, and by the private sector. These long-term power purchase agreements cover 90% of wholesale power volumes in India.<sup>25</sup> With some regional and national coordination, states use a combination of economic dispatch and historical practices (heuristics) to choose which plants to procure power from to deliver to consumers.<sup>25,26</sup> Moreover, the central government and state governments have overlapping jurisdiction over the power sector. Thus, not only do fuel mixes and emissions of electricity consumed vary by state, but so do policies governing emissions from the power sector.<sup>25,27</sup> Lastly, on average, about 20% of electricity generated is lost due to technical and nontechnical losses in transmission and distribution (T&D losses), with values ranging between 9 and 47% in different states.<sup>28,29</sup> To meet a certain level of electricity demand power plants must generate additional electricity, resulting in more emissions.

While it is possible to characterize the emissions of the Indian power sector spatially and temporally based on generation location using daily reported power generation,<sup>30</sup> finer subdaily scale consumption-based estimates require some account of power transfers between states, diurnal patterns in power demand, an accounting of both T&D losses, and any demand not met due to rolling blackouts, i.e., loadshedding.<sup>31</sup> Moreover, currently reported data can only provide estimates of average emission factors, i.e., total emissions divided by total generation. However, average emissions factors may be a poor representation of the changes in emissions that arise from interventions in the power sector that would lead to small changes in electricity demand.<sup>32,33</sup> Instead, marginal emission factors (MEF) are likely to provide a better representation. These MEFs estimate the change in emissions from marginal generation, i.e., the electricity generation that increases or decreases to meet a change in demand. These factors provide more accurate assessments especially when analysts or policymakers evaluate interventions that cause changes in electricity demand likely to vary seasonally or time-of-day, such as electric vehicles, air conditioning, or energy efficiency. Estimating MEFs require subdaily accounting of diurnal demand and generation patterns because different mixes of generators and fuels may respond to changes in demand depending on season and time-of-day.<sup>34</sup>

Dispatch or production-cost models provide one way to address current limitations to characterize Indian power sector emissions factors. These models simulate scheduling and dispatch of power generators at finer hourly or minute resolution based on power demand.<sup>35,36</sup> Sengupta et al.<sup>36</sup> use a reduced-form dispatch model to derive production and consumption-based average emission factors for CO<sub>2</sub> and SO<sub>2</sub> from Indian power generation. That paper finds that heterogeneity in contracted capacity mixes between states results in differences of up to 26% in average CO<sub>2</sub> emission factors for certain Indian states than estimated by a single national emission factor. However, they do not quantify MEFs.

Previous work has quantified MEFs in the United States,<sup>33,34,37–39</sup> United Kingdom,<sup>32</sup> Canada,<sup>40</sup> and China,<sup>41</sup> but detailed estimates for India remain absent. The only estimate available for India is a single, national operating margin emission estimate for CO<sub>2</sub><sup>42</sup> developed as part of the Clean Development Mechanism (CDM) of the United Nations Framework Convention on Climate Change

(UNFCCC). The CDM is a way for developed countries to offset emissions in developing countries. This value represents the displaced CO<sub>2</sub> emissions for any CDM infrastructure projects, and the United Nations and other international organizations use this value to calculate resulting carbon offsets.<sup>43</sup> Current estimates used in analysis do not adequately reflect subnational variability in marginal emissions from power sector interventions in India. This becomes increasingly relevant as countries around the world, including India, set net-zero emissions targets, and climate finance funds climate mitigation efforts.

Here, we use scheduling and dispatch (production cost) model to quantify spatially, temporally, and seasonally the MEFs for CO<sub>2</sub>, SO<sub>2</sub>, NO<sub>x</sub>, and primary (directly emitted) PM<sub>2.5</sub> under current (2017–2018) and future (2030–2031) grid conditions in India. Our analysis is the first analysis to characterize a growing and changing Indian power sector using spatially and temporally variable MEFs. They provide policymakers, analysts, and researchers emission factors more representative of Indian power system operations.

## MATERIALS AND METHODS

### Reduced-Form and Full-Form Model Descriptions.

We simulate CO<sub>2</sub>, SO<sub>2</sub>, NO<sub>x</sub>, and PM<sub>2.5</sub> MEFs using two sets of power dispatch models: (1) a reduced-form model<sup>36,44</sup> developed for Indian power generation to simulate 2017–2018<sup>36</sup> and (2) a full-form model<sup>45</sup> to simulate 2030–2031. We briefly describe the models here, but we direct readers to cited papers for further details. The 2017–2018 reduced-form model creates a supply curve of available generating capacity by first ordering generating units by a variable cost of power (such as fuel and other VO&M).<sup>46–49</sup> It then takes net demand for the hour (electricity demand minus renewable generation) and chooses generators whose cumulative sum based on the supply curve equals net demand. This is equivalent to minimizing the hourly total system cost of electricity to meet net demand. This method assumes no other constraints or costs including those from transmission, generator ramp limits, or startup costs. When examined *ex post*, this assumption has limited impacts on both model estimates of generation from individual power units (<10%) and aggregate system generation estimates (<1%).

The reduced-form model simulates each hour between September 2017 to August 2018 using all dispatchable generators (coal, gas, nuclear, and hydro) based on inventories of power plants in India from daily generation reports with outages.<sup>30</sup> Table S1 shows the fuel-wise capacity incorporated in the reduced-form model.<sup>36</sup> The structure of the model is individual models for each Indian state to mirror the decentralized manner of Indian power dispatch. Each state self-schedules generators to meet the demand based on a portfolio of long-term power purchase agreements (PPA) entered by the state. The reduced-form model incorporates this reported capacity tied to PPAs,<sup>50,51</sup> which govern 90%+ generation in India.<sup>26,52,53</sup> Note that the 2017–2018 reduced-form model does not include all installed capacity but only those modeled from reported capacity in long-term contracts, approximately 75–85% of all installed capacity (India has a current capacity surplus relative to peak demand).<sup>50</sup> Likewise, it allocates intrastate capacity (owned by state governments) 100% to each respective state, and interstate capacity (owned by central government and private sector) to beneficiary states by allocating fractions of capacity <100% consistent with

current practices.<sup>51</sup> To reflect seasonal constraints on hydro, the model constrains hydrocapacity available in each hour using reported hydrogeneration by plant<sup>30</sup> and diurnal hydrogeneration profiles for each month.<sup>54</sup>

The model presented by Sengupta et al (2022)<sup>36</sup> only simulates each individual state, implicitly modeling interstate electricity transfers through fractional allocations of interstate generation capacity. This method results in approximately 3% annual national demand (varying by hour and state) not met due to idle capacity. Idle capacity is contracted capacity in state not being available to an outside state should the model determine it is not needed to meet demand within the state. For example, if 60% of a 500 MW unit is only needed to satisfy the demand in a contracted state, the remaining 40% (200 MW) would not generate. This 40% would likely participate in bilateral exchanges with other states or short-term power market.<sup>26,53</sup> To simulate this interstate exchange of power and to meet the 3% shortfall in annual demand, we rerun the reduced-form model each hour at regional and national levels (see Figure S25 for a map of power regions in India) by ordering leftover capacity by variable cost and dispatching accordingly to meet unmet net demand. This procedure results in 4 of 8 592 simulated (7 days throughout the year do not have reported demand data) with unmet net demand anywhere in the country. The highest deficit in generation among these hours was 1.0 GWh, easily met by capacity not incorporated into the model.

Inputs for demand come from reported daily demand<sup>31</sup> disaggregated by diurnal monthly demand profiles.<sup>55</sup> The demand values we use represent energy met injected at the border of each Indian state, so they are net values that account for both load shedding and T&D losses. We pair these demand values with monthly reported renewable generation for each state<sup>56</sup> disaggregated by diurnal renewable generation profiles for each month<sup>54</sup> to estimate net demand.

To estimate emissions associated with simulated generation, we capture the heterogeneity in emission factors for each generating unit. The reduced-form model uses CO<sub>2</sub>, SO<sub>2</sub>, NO<sub>x</sub> and PM<sub>2.5</sub> emission factors for each generating unit estimated by heat rate<sup>18,57</sup> and weighted-average Indian coal composition.<sup>58,59</sup> We assume no sulfur or NO<sub>x</sub> control at the coal power stations but 90% control of PM<sub>2.5</sub> emissions. Current Indian regulation stipulates installation of SO<sub>2</sub> and NO<sub>x</sub> controls.<sup>60</sup> However, approximately 5% of plants have installed control equipment as of December 2019<sup>61</sup> with the Government of India extending deadlines to install equipment to 2025.<sup>60</sup> Moreover, the lack of monitoring<sup>24</sup> and enforcement means it is difficult to verify whether installed control equipment runs regularly.<sup>62</sup> This approach captures the variation in emission rates for each generating unit.<sup>23</sup> For gas plants, we assume zero sulfur and PM<sub>2.5</sub> emissions and a constant NO<sub>x</sub> emission factor of 2.95 kg/MWh from U.S. EPA (2018).<sup>63</sup>

When evaluating the reduced-form model, model reproduces reported values well, which we briefly discuss here, and we refer the reader to the Supporting Information for a detailed comparison of reduced-form results versus reported data. Evaluating the model's estimates of hourly generation against hourly reported nationwide generation for 2018–2019, we find an annual correlation of 0.82 (Figure S1a). When comparing estimated generation to reported generation by region, fuel, and month, the model estimates a root-mean-square error (RMSE) of 1 TWh and mean absolute error (MAE) of 0.49

TWh, with the highest error for gas generation in northern and southern regions (Figure S2). Lastly, we find a RMSE of 0.16 and MAE of 0.11 when comparing the estimated and reported annual fractions of out-of-state generation for each state (Figure S3). Here, both RMSE and MAE values closer to zero indicate better agreement.

The 2030–2031 full-form model<sup>45,64</sup> uses open-source Python for Power System Analysis (PyPSA)<sup>65</sup> customized for India, PyPSA-India.<sup>45</sup> The configuration of PyPSA-India is a unit commitment and dispatch model. Each state is treated as a balancing area node connected by interstate transmission capacity, and the model finds the combination of generators that minimizes total system costs to meet demand in an hour for each state. There are additional constraints and costs on generator minima, ramp rates, startup, and transmission included in this optimization. Additionally, PyPSA-India accounts for generator outages probabilistically by defining the likelihood a generator is out in each hour, based on historical data and interviews with Indian power sector experts.

The full-form model simulates April 2030 to March 2031 generator capacity mixes exogenously defined by the National Electricity Plan developed by the Government of India.<sup>66</sup> These generators have marginal costs from similar reported and estimated sources as the reduced-form model.<sup>50</sup> The capacity mixes shown in Table S1 show differences in the full-form model from the reduced-form model in thermal capacity (50%) as well as differences in hydro (100%) and renewables (380%) in line with Government of India targets to expand nonfossil capacity by 2030.<sup>16</sup> The model assumes integrated markets, i.e., capacity anywhere in the country can meet demand anywhere assuming sufficient transmission. This contrasts with the fragmented, decentralized way the Indian power system currently operates and incorporated into the reduced-form model. Like the reduced-form model, the model enforces daily constraints on hydrogeneration to reflect seasonal water availability. To produce renewable generation inputs, the full-form model first allocates renewable generation capacity to states based on resource potential, land-use, and infrastructure. Then using the System Advisor Model from the National Renewable Energy Laboratory,<sup>67</sup> the authors derive hourly renewable generation profiles.

To derive demand estimates for 2030–2031, Spencer et al. (2020)<sup>45</sup> assume hourly load profiles shapes the same as from 2008 to 2018 and continuation of historical growth rates, with some minor adjustments. Consequently, when comparing input demand profiles between the reduced-form 2017–2018 model and the full-form 2030–2031 (Figure S4), we find for most states and seasons, hourly load profiles between the two time periods are highly correlated (>0.8). Moreover, these 2030–2031 demand assumptions are net values which account for both load shedding and T&D losses.

For each generating unit in the full-form model, we estimate unit-specific emission factors using similar methods as the reduced-form 2017–2018 model,<sup>59</sup> in many cases using the same emission factors for units existing through 2030–2031. For newer units, we estimated heat rates using fits based on unit capacity size. Like 2017–2018, we assume no SO<sub>2</sub> or NO<sub>x</sub> control and 90% control of PM<sub>2.5</sub> emissions.

To explicitly estimate the shares of each generator to each state's demand for the 2030–2031 model, we use the “downstream-looking” power tracing algorithm presented by Bialek et al. (1996).<sup>68</sup> The 2017–2018 reduced-form model explicitly estimates each hour the share of each generator unit

supplying electricity to meet demand in each state. The 2030–2031 full-form model on the other hand balances power in each state as a node and simulates simultaneous power flows between states in its optimization to determine which units generate to meet demand at each state. The algorithm of Bialek et al. (1996)<sup>68</sup> assumes perfectly mixed nodes where power flow out of a state is proportional to power flow into a state from respective sources. It traces power flow for each estimated hour to apportion generation and emissions from each generating unit to each state's demand. This results in an explicit time series of generation and emissions tied to generating unit for each state. We present the fractions of in-state and out-of-state electricity estimated by the 2030–2031 full-form model in Figure S5 which shows the results of this power tracing algorithm.

**Estimating Marginal Emission Factors and Marginal Fuel Shares.** We calculate MEFs for each state using two methods: linear regression<sup>32–34,37,39</sup> and differentiation of emissions-generation curves.<sup>38</sup> We do not estimate MEFs by identifying the emission factor of the marginal generator (i.e., the last, costliest generator in the merit order) because (1) more than one generator or combinations of generators may change output to meet demand and (2) grid operators in India use hydrogenerators as marginal generators<sup>25</sup> based on nonmerit order heuristics, and (3) we have access to only generation and transmission output in the 2030–2031 modeling results, not the merit order input. However, as sensitivity, we present emission factors from the marginal generator in 2017–2018 in Figures S26–S35. To account for the load-following nature of hydro<sup>36</sup> in the reduced-form model, we keep hydro at the beginning of the merit order but constrain its available capacity throughout the day using reported hydro generation.<sup>30,69</sup> These modeling assumptions account for the times-of-day and seasons when hydro-generators may form marginal generation.

Each annual model simulation produces approximately 8760 h of emissions, generation to meet demand, and fuel shares for each state. We partition the data for each state into two seasons: monsoon (June–October) and dry (January–May; November–December). For each season, we further divide the data into five demand periods based on time-of-day: morning peak (6:00 AM to 10:00 AM), midday (10:00 AM to 2:00 PM), afternoon (2:00 PM to 6:00 PM), evening peak (6:00 PM to 10:00 PM), and overnight (10:00 PM to 6:00 AM). We define hours in each period based on visual inspection of the average diurnal national demand profile in each season.<sup>54,69</sup>

To calculate marginal emission factors using linear regression, we first calculate the hour-to-hour (h) change in emissions ( $E$ ) and generation ( $G$ ) associated with meeting demand for each state ( $s$ ) predicted by the dispatch model. We then partition the changes by demand period ( $p$ ):

$$\Delta E_{s,p,h} = E_{s,p,h} - E_{s,p,h-1} \text{ (kg)} \quad (1)$$

$$\Delta G_{s,p,h} = G_{s,p,h} - G_{s,p,h-1} \text{ (MWh)} \quad (2)$$

Here  $\Delta E_{s,p,h}$  and  $\Delta G_{s,p,h}$  are marginal emissions and marginal generation, respectively. Because generation meets demand at the border of a state (net loadshedding and T&D losses), it is also representative of power consumption. The marginal emission factor ( $\beta_{s,p}$ ) in kg/MWh for each state in each demand period takes the functional form:

$$\Delta E_{s,p,h} = \beta_{s,p} \Delta G_{s,p,h} + \alpha_{s,p} + \varepsilon_{s,p} \quad (3)$$

With  $\alpha_{s,p}$  and  $\varepsilon_{s,p}$  as the regression intercept and error term, respectively.

We have 10 marginal emission factors for each state: five for each season (monsoon and dry). We also calculate MEFs using differentiation of total emission-total generation curves. While regression-based marginal emission factors show average changes in emissions per unit change in demand for specified time periods, differentiation-based MEFs can show MEFs as a continuous function of demand. We use the method described by Deetjen and Azevedo (2019).<sup>38</sup> First, we order by increasing demand ( $d_{s,n}$ ), the hourly time series of emissions ( $E_{s,h}$ ), and generation ( $G_{s,h}$ ) for each state with  $n$  data points. Then, we take rolling means of emissions and generation with 5% subsets of the data. This constructs the total emissions ( $E_{s,\text{total}}$ )–total generation ( $G_{s,\text{total}}$ ) curve for each state. Then we calculate difference quotients of this curve as an approximation of the curve's derivative to estimate MEF as a function of demand:

$$\text{MEF}(d_{s,n}) = \frac{E_{s,\text{total}}(d_{s,n}) - E_{s,\text{total}}(d_{s,n-1})}{G_{s,\text{total}}(d_{s,n}) - G_{s,\text{total}}(d_{s,n-1})} \quad (4)$$

To construct the MEF-demand curve, we calculate rolling means with 5% subsets of the MEFs calculated by difference quotients. Note  $G_{s,\text{total}} = d_{s,n}$  because generation meets all demand as our demand estimates account for loadshedding and T&D losses.

To calculate the fractional share of marginal generation by various fuel types (e.g., coal, hydro, renewables, etc.), we define the hourly change in generation from each fuel type,  $f$  in each period  $p$ :

$$\Delta G_{s,p,h,f} = G_{s,p,h,f} - G_{s,p,h-1,f} \quad (5)$$

Analogous to marginal emissions, the marginal fuel share ( $\gamma_{s,p,f}$ ) from fuel  $f$  takes the functional form:

$$\Delta G_{s,p,h,f} = \gamma_{s,p,f} \Delta G_{s,h} + \alpha_{s,p,f} + \varepsilon_{s,p,f} \quad (6)$$

and  $\alpha_{s,p,f}$  and  $\varepsilon_{s,p,f}$  are the regression intercept and error term, respectively. Marginal fuel share quantifies the frequency each fuel type meets marginal generation. Because marginal generation is met by the sum of marginal generation from each fuel type, this implies

$$\sum_f \Delta G_{s,p,h,f} = \Delta G_{s,p,h} \quad (7)$$

$$\sum_f \gamma_{s,p,f} = 1 \quad (8)$$

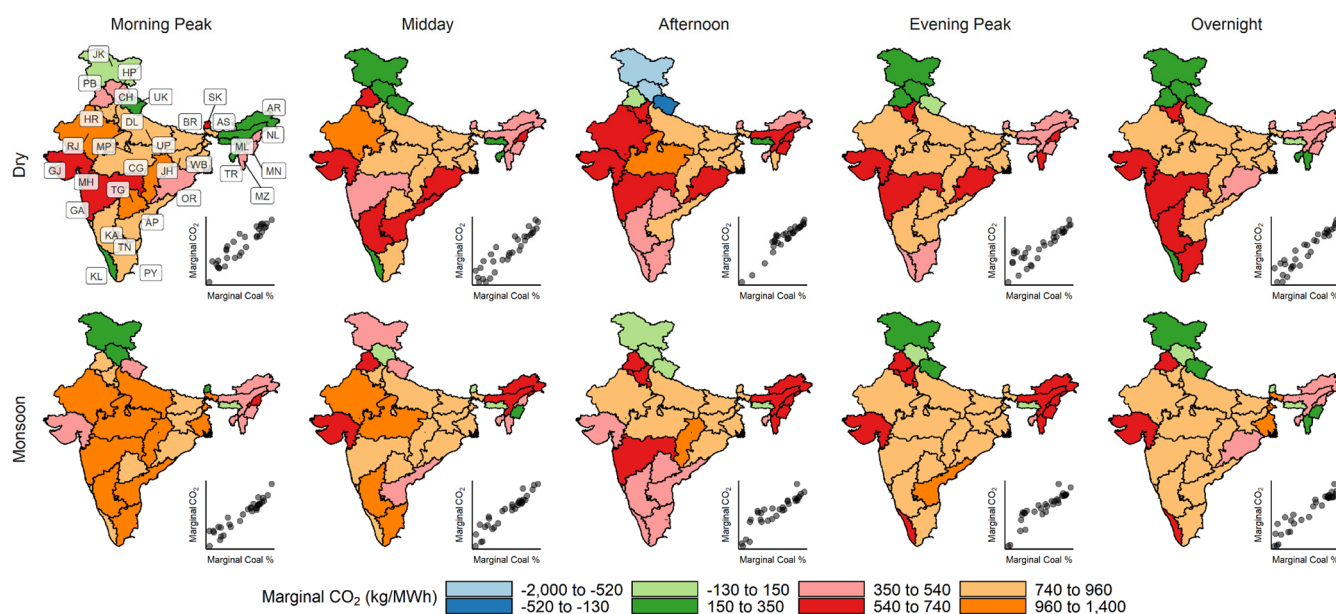
Similarly, we estimate the share of marginal generation of each fuel type  $MS_f$  using similar methods. Instead of emissions, we take rolling means of generation from each fuel type,  $G_{s,f,\text{total}}$ :

$$MS_f(d_{s,n}) = \frac{G_{s,f,\text{total}}(d_{s,n}) - G_{s,f,\text{total}}(d_{s,n-1})}{G_{s,\text{total}}(d_{s,n}) - G_{s,\text{total}}(d_{s,n-1})} \quad (9)$$

$$\sum_f MS_f(d_{s,n}) = 1 \quad (10)$$

## RESULTS

We focus our presentation on marginal CO<sub>2</sub> emission factors because emissions of all pollutants (SO<sub>2</sub>, NO<sub>x</sub>, PM<sub>2.5</sub>, and



**Figure 1.** 2017–2018 regression-based, marginal CO<sub>2</sub> emission factors for dry (top) and monsoon (bottom) seasons mapped for each Indian state. Columns separate the time-of-day in each season. Scatter plots inset show the strong correlation between average percentage of marginal coal generation and marginal CO<sub>2</sub> emission factors for a state. Each point represents one state in each season and time-of-day.

CO<sub>2</sub>) from individual generation units scale with fuel throughput and efficiency. Therefore, we largely see the same trends in marginal CO<sub>2</sub> emission factors as we do for other pollutants especially because emission of other pollutants scales with coal generation. We present marginal emission factors for other pollutants in the Supporting Information (Figures S8–S13).

In Figure 1, we show the time-of-day resolved marginal CO<sub>2</sub> emission factors mapped for electricity consumed in each Indian state based on the 2017–2018 reduced form dispatch model.<sup>36</sup> Each column shows the time-of-day period, the top row shows emission factors for dry season, and the bottom row shows emission factors for the monsoon season. The first panel shows abbreviations for each Indian state. In these maps, negative marginal emission factors mean emissions *decrease* with increases in marginal power demand due to greater shares of lower emission generating sources meeting new demand. For example, if shares of renewable generation increase to meet increases in demand, all else equal, the overall average marginal change in emissions would be negative, inducing a negative marginal emission factor.

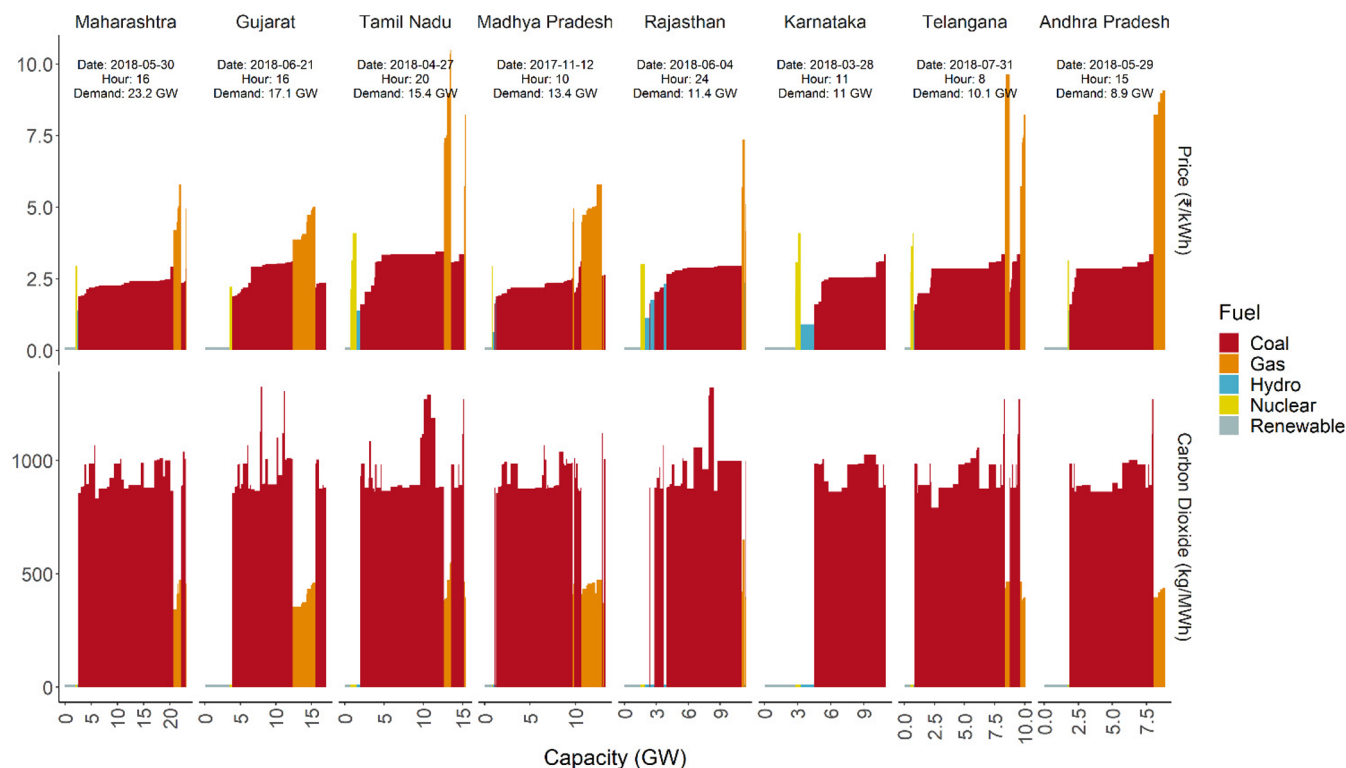
Our estimated marginal emission factors for CO<sub>2</sub> span 3 orders of magnitude across 32 states in Figure 1, and the nationwide average CO<sub>2</sub> emission factor of 711 kg/MWh does not capture this heterogeneity. The fractional share of marginal generation by various fuel types in each state, season, and time-of-day (eq 6) explain the differences we see in CO<sub>2</sub> MEFs, with the average percentage shares of marginal coal generation highly correlated with CO<sub>2</sub> MEFs. Each panel of Figure 1 also has an inset that shows this strong correlation. In these inset plots, the *x*-axis shows the average percentage of marginal coal generation calculated for each state and time-of-day period using eq 6, and the *y*-axis shows the corresponding CO<sub>2</sub> MEF mapped. Each point represents a single state for each time-of-day and season. We show the fractional shares of marginal generation by all fuel types in each state, season, and time-of-day in Figures S14 and 15. Likewise, Tables S2 and S3 show the estimates mapped in Figure 1 with regression standard

error and coefficients, and Figure S6 shows the estimates of Figure 1 plotted on a linear scale for another point of comparison.

Overall, two broad groups of states emerge in both seasons and all periods. The first are 16 higher emission states with higher electricity demand located throughout the country: Puducherry (PY), Chhattisgarh (CG), Rajasthan (RJ), Madhya Pradesh (MP), Bihar (BR), West Bengal (WB), Jharkhand (JH), Goa (GA), Uttar Pradesh (UP), Telangana (TG), Karnataka (KA), Tamil Nadu (TN), Delhi (DL), Andhra Pradesh (AP), Haryana (HR), and Maharashtra (MH). The second are 16 lower emission states with lower electricity demand located mostly in hydro-rich northern and northeastern Himalayan areas: Odisha (OR), Nagaland (NL), Gujarat (GJ), Punjab (PB), Mizoram (MZ), Kerala (KL), Arunachal Pradesh (AR), Assam (AS), Manipur (MN), Sikkim (SK), Chandigarh (CH), Tripura (TR), Uttarakhand (UK), Jammu and Kashmir (JK), Meghalaya (ML), and Himachal Pradesh (HP). There are some exceptions to these general groupings, such as Gujarat and Punjab, which have lower emissions but high demand and Puducherry and Goa which have high emissions and low demand.

For higher emission states, CO<sub>2</sub> MEFs have a median 860 kg/MWh across all seasons and times-of-day which we see with the widespread orange shading in Figure 1. In these states, coal dominates marginal generation, which drives higher emissions. During the dry season, median coal marginal generation varies between 75% (afternoon) and 87% (morning peak) of the time across these states (Figure S14), with similar patterns during the monsoon season (Figure S15). Due to high marginal coal generation, marginal emissions are tightly correlated with marginal generation, and the median *r*<sup>2</sup> value for CO<sub>2</sub> MEFs being 0.92 among all seasons and times-of-day (Tables S2 and S3).

For lower emission states, CO<sub>2</sub> MEFs have median 460 kg/MWh for all seasons and times-of-day. In these states, we see more hydro and/or gas marginal generation than coal marginal generation driving lower CO<sub>2</sub> MEFs. In these states, median



**Figure 2.** Hourly dispatch (merit) order for maximum hourly demand in eight high renewable states. The top panel shows capacity ordered by variable cost, and the bottom panel shows CO<sub>2</sub> emission factors of the same capacity. Capacity is ordered by renewable capacity and nuclear capacity first, which are must-run. Then hydro, coal, and gas capacities contracted with the state are ordered by variable cost. Lastly, leftover capacity not used by contracted states is ordered at the regional and national levels.

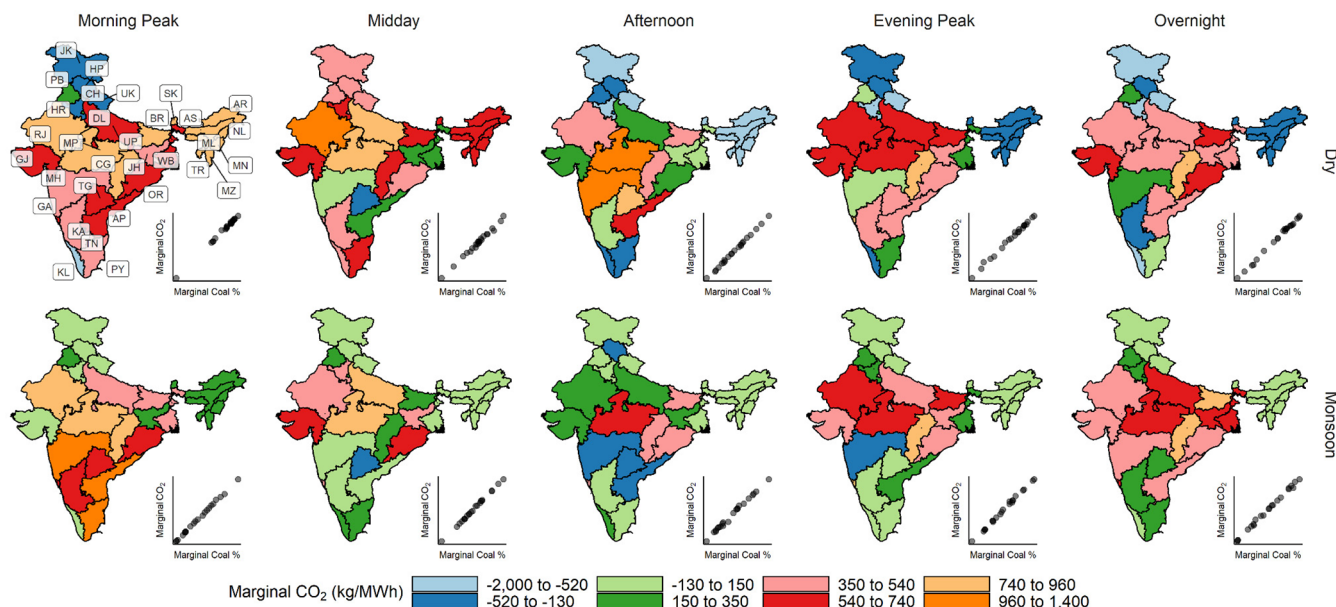
hydro generation ranges between 31% (afternoon) and 47% (overnight), and marginal gas generation ranges between 16% (overnight) and 29% (afternoon) (Figures S14 and S15). Meanwhile the dry season marginal coal generation numbers are between 19% (afternoon) and 27% (evening peak). We find similar trends in the monsoon season (Figure S15) with slightly more marginal coal generation than the dry season. Because of higher amounts of marginal generation from lower-emission or zero-emission sources, marginal emissions are not as correlated with marginal generation. The median  $r^2$  value for CO<sub>2</sub> MEFs in these lower emission states is 0.56 across all states and times-of-day (Tables S2 and S3).

In general, we see little seasonal temporal variability in CO<sub>2</sub> MEFs between monsoon and dry seasons with monsoon MEFs slightly higher than dry MEFs due to the dominance of marginal coal generation. Moreover, lower emission states show more seasonal variability than higher emission states due to higher marginal hydro generation in these states (Figures S14 and S15). Seasonally, across all times of the day, the median absolute difference between monsoon and dry CO<sub>2</sub> MEFs in 16 lower emission states is 140 kg/MWh. The same value is 100 kg/MWh for the 16 higher emission states. Moreover, in these higher emission states, median time-of-day marginal hydro generation is higher during the dry season (2–14%) than the monsoon season (3–9%), consistent with Indian dispatch practices that use hydro as marginal generators during dryer months<sup>25</sup>

Regarding intraday temporal variability, we see lower levels of variability between times-of-day in both high emission and low emission states across both seasons. The monsoon season sees more time-of-day variability than the dry season. Changes

in marginal renewable generation along with changes in demand largely induce these variations (Figures S14 and S15). Across low emission states, the median time-of-day CO<sub>2</sub> MEF ranges from 420 kg/MWh (overnight) to 620 kg/MWh (evening peak) during the monsoon season and 350 kg/MWh (overnight) to 460 kg/MWh (afternoon) during the dry season. Likewise, across high emission states, the median time-of-day CO<sub>2</sub> MEF in ranges from 800 kg/MWh (afternoon) to 980 kg/MWh (morning peak) during the monsoon season and 800 kg/MWh (overnight) to 890 kg/MWh (morning peak) during the dry season.

In Figure 2, we show the dispatch (merit) order of capacity during the annual maximum hourly demand for eight high renewable states from the 2017–2018 reduced-form model: Maharashtra, Gujarat, Tamil Nadu, Madhya Pradesh, Rajasthan, Karnataka, Telangana, and Andhra Pradesh. The top panel shows the capacity ordered by price (variable cost) in Indian rupees per kWh (Rs/kWh), and the bottom panel shows the emissions factors of that ordered capacity in kg/MWh. We show the annual maximum hourly demand to show the high capacity amounts available to meet demand in each state. We order the capacity by price at three levels: state, regional, and national to reflect the three levels of successive dispatch the reduced form dispatch model assumes (see Methods, Figure S25). Overall coal dominates dispatch in these states, despite high renewable capacity, especially for states where maximum annual demand occurs at during midday or afternoon hours when renewable capacity available is the highest (for Karnataka and Andhra Pradesh). A caveat with these dispatch orders is that renewable and hydro capacity may change from hour to hour and thus contribute to meeting



**Figure 3.** 2030–2031 regression-based, marginal CO<sub>2</sub> emission factors for the dry (top) and monsoon (bottom) seasons mapped for each Indian state. Columns separate the time-of-day in each season. Scatter plots inset show the strong correlation between average percentage of marginal coal generation and marginal CO<sub>2</sub> emission factors. Each point represents one state in each season and time-of-day.

changes in demand. The frequency of marginal renewable or hydro generation can be positive or negative during these hours (Figures S14 and S15). Thus, more than one fuel type may meet marginal demand and determine marginal CO<sub>2</sub> emissions. Nonetheless, because coal capacity dominates dispatch order in these states as it does in most states (>50%), it meets marginal demand >50% of the time (Figures S14 and S15).

In Figure 3, we show the mapped time-of-day resolved marginal CO<sub>2</sub> emission for electricity consumed in each Indian state based on the 2030–2031 full form dispatch model.<sup>45</sup> Figure 3 is analogous to Figure 1: rows show each season, columns show each time-of-day, and each panel has an inset showing the strong correlation between average percentage shares of marginal coal generation (eq 6) and CO<sub>2</sub> MEFs. Likewise, we show the fractional shares of marginal generation by all fuel types in each state, season, and time-of-day in Figures S16 and S17. Tables S10 and S11 show the estimates mapped in Figure 3 with regression standard error and coefficients, and Figure S7 shows the estimates of Figure 1 plotted on a linear scale for another point of comparison. Future 2030–2031 simulations treat the low-demand, north-eastern states of Sikkim (SK), Assam (AS), Mizoram (MZ), Tripura (TR), Arunachal Pradesh (AR), Nagaland (NL), Manipur (MN), and Meghalaya (ML) as one Northeast state. Likewise, these simulations treat territories of Chandigarh (CH) in north India and Puducherry (PY) in south India as part of the neighboring states of Punjab (PB) and Tamil Nadu (TN), respectively.<sup>45</sup> To simplify comparisons between 2030 and 2031 and 2017–2018 estimates, we ignore these territories and Northeastern states, which form 1–2% of annual national demand during respective simulation periods.

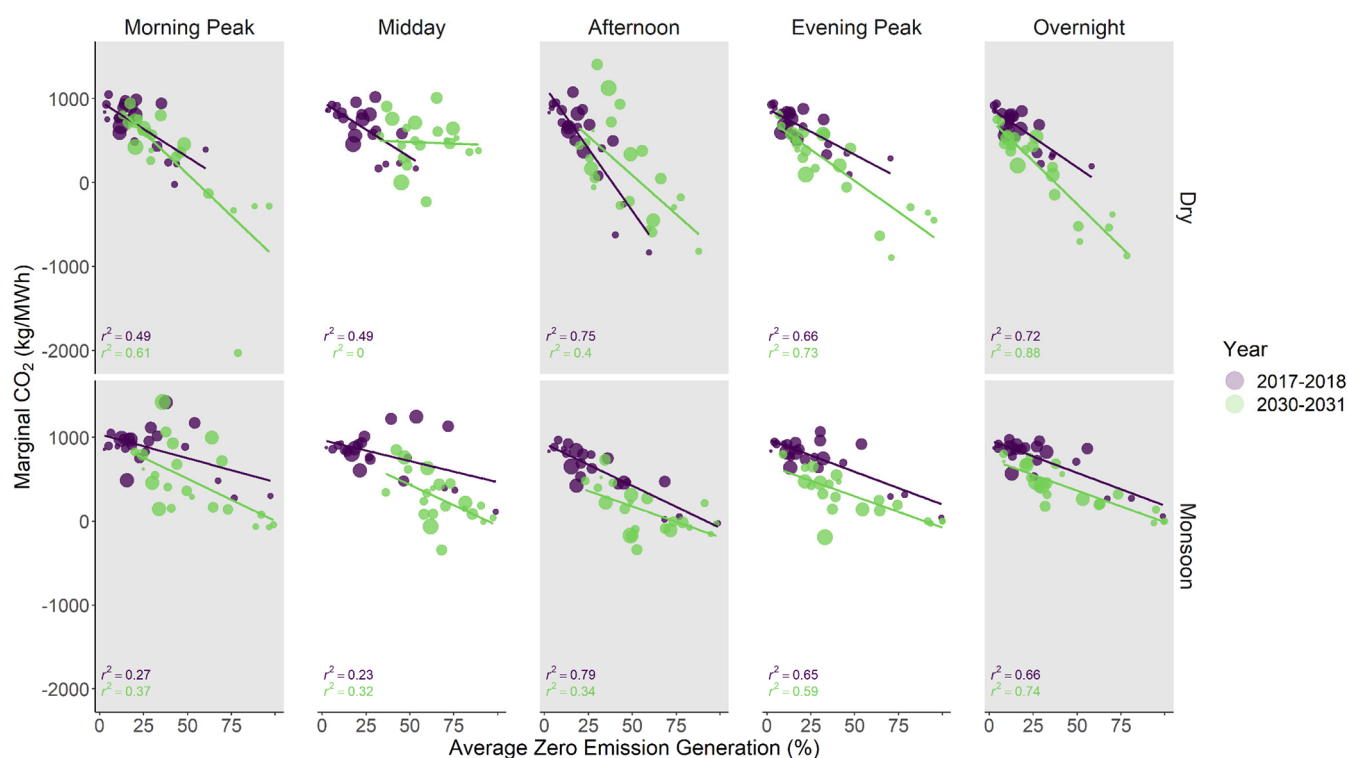
Figure 3 shows that under 2030–2031 assumptions, estimated marginal CO<sub>2</sub> emission factors are generally lower than those in 2017–2018 but show similar variability between states. Emission factors once again vary across 3 orders of magnitude between states, with the same 16 higher emission, higher demand states in 2017–2018 (Chhattisgarh [CG],

Rajasthan [RJ], Madhya Pradesh [MP], Bihar [BR], West Bengal [WB], Jharkhand [JH], Goa [GA], Uttar Pradesh [UP], Telangana [TG], Karnataka [KA], Tamil Nadu/Puducherry [TN/Puducherry], Delhi [DL], Andhra Pradesh [AP], Haryana [HR], and Maharashtra [MH]) having relatively higher emission factors as 2017–2018. In these states, 2030–2031 CO<sub>2</sub> MEFs have median 420 kg/MWh among all seasons and times-of-day. In these states, decreases in marginal coal generation and increases in marginal hydro and renewable generation explain this decrease in CO<sub>2</sub> MEF relative to 2017–2018. Median marginal coal generation across all seasons and times-of-day in these higher emissions states decreases to 43%. Median correlation coefficients  $r^2$  decrease to 0.17 as well, showing the weaker relationship between larger shares of zero-emission generation and emission.

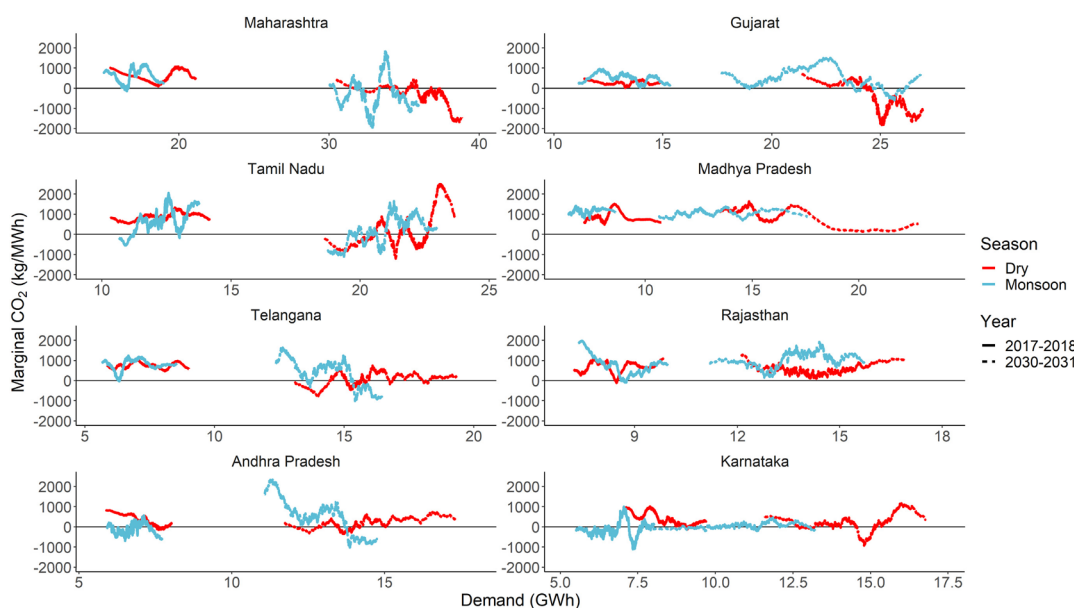
For the remaining lower emission, lower demand states in 2030–2031 marginal CO<sub>2</sub> MEFs decrease further with a median of 160 kg/MWh. Like higher emission states, decreases in marginal fossil generation, and increases in marginal hydro and renewable generation explain this decrease. Median marginal gas generation decreases to 0.03%, and median coal generation decreases to 8%. Due to the very weak relations between marginal generation and marginal emissions, the median  $r^2$  value decreases to 0.05.

Furthermore, we see more seasonal variability in both high emission and low emission states in 2030–2031 largely due to the increases in marginal renewable generation and marginal hydrogeneration, respectively (Figures S16 and S17). Like 2017–2018, lower emission states show more difference between the dry and monsoon seasons than higher emission states, consistent with higher marginal hydro generation. The median absolute seasonal difference in CO<sub>2</sub> MEFs across all times-of-day in lower emission states increases to 320 kg/MWh in 2030–2031. For higher emission states, this median value increases to 150 kg/MWh.

Under 2030–2031 assumptions, we also see more temporal variability between times-of-day in CO<sub>2</sub> MEFs. In general, this is due to increased marginal generation from renewable



**Figure 4.** Marginal CO<sub>2</sub> emissions versus average zero emission (hydro, renewables, and nuclear) generation in each demand period in the dry and monsoon seasons for 2017–2018 and 2030–2031. Each dot represents one state and the size of dots scales to annual demand for each state. The lines are lines of best fit.



**Figure 5.** The 2017–2018 and 2030–2031 dry and monsoon season marginal CO<sub>2</sub> emission factors as a function of demand in eight high variable renewable energy (solar, wind) states. In 2017–2018, these eight states collectively form 85% of all variable renewable energy generation. In 2030–2031, these eight states collectively form 60% of all variable renewable energy generation.

sources, mostly solar (Figures S16 and S17) during midday or afternoon hours decreasing CO<sub>2</sub> MEFs. Meanwhile marginal coal generation increases during morning peak, evening peak, and overnight hours especially in high emission states increasing CO<sub>2</sub> MEFs. Across high emission states, the median time-of-day CO<sub>2</sub> MEF in ranges from 150 kg/MWh (afternoon) to 680 kg/MWh (morning peak) during the monsoon season and 380 kg/MWh (afternoon) to 590 kg/MWh

(morning peak) during the dry season. Furthermore, in these states, we see disparities in MEFs during midday and afternoon hours when renewable generation peaks. For example, during afternoon hours in both seasons, we see lower MEFs in the southern states but higher coal-dominated MEFs in the eastern states. In low emission states, the median time-of-day CO<sub>2</sub> MEF in both seasons ranges from 20 kg/MWh (afternoon) to 160 kg/MWh (midday) during the monsoon season and –330

kg/MWh (evening peak) to 470 kg/MWh (midday) during the dry season.

There is a 1:1 relation between decreasing CO<sub>2</sub> MEFs and increasing zero-emission marginal generation, i.e., increased hourly generation to meet the new hourly load. However, the relationship between increased average zero-emission generation and CO<sub>2</sub> MEFs is not as direct because average nonemitting generation may not change when there is an increase or decrease in electricity demand. We explore this relationship in Figure 4.

Figure 4 shows marginal CO<sub>2</sub> emissions in each demand period against the average percentage of zero emission generation (renewables, hydro and nuclear), in the dry and monsoon seasons, for both 2017–2018 and 2030–2031. Each point represents a state, and the size of the point scales to the annual demand in each simulation period. The lines are lines of best fit.

In the dry season in 2017–2018 (violet points, top panel Figure 4), average zero-emission generation shares and CO<sub>2</sub> MEFs are the most correlated during the afternoon and overnight periods with  $r^2$  values of 0.75 and 0.72, respectively. During evening peak periods, correlation decreases to  $r^2 = 0.66$ . The relationship weakens the most during morning peak and midday periods when  $r^2$  values decrease to 0.49 in both periods.

In the 2030–2031 dry season, during all periods except midday afternoon, the relationship strengthens (green points, top panel Figure 4). Increasing average zero emission generation is associated with decreasing CO<sub>2</sub> MEFs. During morning peak, evening peak, and overnight hours,  $r^2$  values increase to 0.61, 0.73, and 0.88, respectively. In midday and afternoon hours,  $r^2$  in 2030–2031 however decreases to 0.00 and 0.4, respectively.

In the monsoon season in 2017–2018 (violet points, bottom panel Figure 4), increasing average zero-emission generation is most correlated with decreasing marginal emission factors during afternoon and overnight hours,  $r^2$  of 0.79 and 0.66, respectively. During morning peak, midday, and evening peak hours,  $r^2$  is 0.27, 0.23, and 0.65, respectively.

In the 2030–2031 monsoon season (green points, bottom panel Figure 4), correlation increases in all periods except afternoon and evening peak hours. During morning peak  $r^2 = 0.37$ , during midday  $r^2 = 0.32$ , and during overnight  $r^2 = 0.74$ . However, during afternoon and evening periods correlation decreases,  $r^2 = 0.34$  and 0.59, respectively.

Figure 5 shows CO<sub>2</sub> MEFs as functions of electricity demand for both the dry and monsoon seasons for 2017–2018 and 2030–2031. We use eq 4 to construct these curves by reordering hourly emissions in order of increasing hourly demand and differentiating the emissions-generation curves. We illustrate eight high renewable capacity states which collectively generate 85% of annual renewable generation in 2017–2018 and 60% in 2030–2031. We show similar curves for SO<sub>2</sub>, NO<sub>x</sub>, and PM<sub>2.5</sub> in Figures S22–S24. We order states by decreasing electricity demand. For example, Maharashtra has the highest demand and Andhra Pradesh has the lowest demand. We construct curves for each season (dry and monsoon) and each modeling time period (2017–2018 and 2030–2031). In these states, power demand roughly doubles from 2017–2018 to 2030–2031 as indicated by curves for 2030–2031 sitting to the right of lines for 2017–2018. We show the associated marginal fuel generations as functions of demand for each state in Figures S18–S21.

In 2017–2018, for all states except Maharashtra, Madhya Pradesh, and Karnataka, we see a significant overlap in the electricity demand between seasons, indicating little difference in demand between seasons. For example, in Maharashtra during the dry season, between 14.7 and 18.6 GWh, CO<sub>2</sub> marginal emission factors decrease from 1014 to 99 kg/MWh but then increase at loads greater than 18.6 GWh. For the same demand range in the monsoon period, marginal CO<sub>2</sub> emission factors oscillate between –147 and 1240 kg/MWh. In general, when comparing these CO<sub>2</sub> MEFs to those estimated through regression earlier, we see wider ranges estimated with CO<sub>2</sub> MEFs from generation-emissions curves than we see with regression-based CO<sub>2</sub> MEFs. Again, taking Maharashtra as an example, CO<sub>2</sub> MEFs as a function of demand range between 99 kg/MWh to 1080 kg/MWh during the dry season while regression-based estimates range between 458 kg/MWh and 683 kg/MWh. In these high renewable energy states, these wider ranges stem from associations between swings in demand and swings in renewable generation (Figures S18 and S19). For a given season, higher levels of marginal renewable generation are associated with certain levels of demand given the demand profiles, e.g., more solar to meet the midday demand. As both demand and renewable generation fluctuate throughout the day, marginal generation from other fuels must increase or decrease, ultimately influencing marginal emissions.

In 2030–2031, Gujarat, Karnataka, and Madhya Pradesh show seasonal differences in demand with less overlap in demand between curves for each season. We see more intraseasonal and interseasonal variation in CO<sub>2</sub> MEFs in all seasons in 2030–2031 compared to 2017–2018 due to further increases in marginal renewable generation and decreases in marginal coal generation (Figures S19 and S20). For example, in the Gujarat dry season, marginal CO<sub>2</sub> emissions vary between –1880 and 701 kg/MWh between electricity demands 21.4 GWh and 27 GWh. Likewise, in the monsoon season in Gujarat, marginal CO<sub>2</sub> emissions vary between –625 and 1500 kg/MWh, while demand spans 17.7 GWh to 26.9 GWh. For Gujarat, the 2030–2031 dry season regression CO<sub>2</sub> MEFs range 336–714 kg/MWh and the monsoon season range 145–630 kg/MWh. Like 2017–2018, these CO<sub>2</sub> MEFs as a function of demand quantify a wider range than regression-based estimates in these states. Higher amounts of marginal renewable generation and its interplay with other marginal generation sources and demand throughout the day yield more variable MEFs (Figures S19 and S20).

## DISCUSSION AND CONCLUSION

In this analysis, we use power dispatch models to quantify the spatial and temporal patterns of marginal CO<sub>2</sub>, SO<sub>2</sub>, NO<sub>x</sub>, and PM<sub>2.5</sub> emission factors in current and possible future Indian power generation. These emission factors are based on power demand in each state with estimates that account for imports and exports between states, load shedding, and T&D losses. Due to a lack of readily available continuous emissions monitoring data or subdaily, spatially resolved power generation data, it is difficult to quantify emissions at sufficient granularity: from which states, during which seasons, and during which hours of the day Indian power generation emissions change due to changes in demand. Moreover, given increasing shares of renewable energy and increasing power demand from economic growth, no analysis has quantified potential emission factors at sufficient spatial and temporal

granularities of a possible Indian grid of the future. Previous analyses quantify the variability in emissions across generators<sup>23</sup> or average annual state or nationwide emission factors in current and future Indian grids.<sup>70–75</sup> This analysis is especially relevant because policy efforts to decarbonize Indian power generation happen at both the central and state-levels so subnational estimates provide more accurate emission factors than commonly used national estimates. Consequently, our estimates are useful to policymakers, analysts, or researchers who may seek to quantify the emissions impacts of new interventions of electricity demand, e.g., electric vehicles, air conditioning, or energy efficiency. The MEFs presented here are more representative of the various types of generation meeting demand during different seasons and times-of-day in India.

Currently, the Indian grid CO<sub>2</sub> MEFs vary considerably across states. We choose to focus our discussion in the main text on CO<sub>2</sub> MEFs since emissions of all pollutants for each generating unit largely scale with efficiency and fuel composition. Despite renewables and hydro decreasing MEFs in lower emission, lower demand states, coal dominates the marginal generation and emissions in higher emission, higher demand states. Further inspection of the merit order under which we simulate plants to generate shows the dominance of coal at the margin. Even in high renewable energy states, where the merit order would fluctuate due to change renewable and hydrogeneration throughout the day, coal still forms the bulk of the available capacity throughout the day. The dominance of coal marginal generation vis a vis marginal renewable and hydro generation moreover explains similar patterns between CO<sub>2</sub> MEFs and MEFs of SO<sub>2</sub>, NO<sub>x</sub>, and primary PM<sub>2.5</sub> in the states with the highest electricity demands.

Furthermore, while we see little seasonal differences in CO<sub>2</sub> MEFs in the current Indian grid, with slightly increased MEFs during the monsoon season than the dry season that is consistent with the current operation of hydro generating units in India. Because of increased rainfall during the monsoon, operators use hydro generation as the baseload capacity during this season. During the dry season when there are further constraints on the amount of water available to generate electricity, e.g., irrigation, drinking water, etc., hydro units are more likely to act as peaking plants that ramp up and down quickly. Our analysis shows higher marginal hydro generation during the dry season (see [Supporting Information](#)). Thus, hydroplants play a greater role in meeting marginal demand and decrease marginal emissions during dryer months.

Our analysis shows that incorporation of greater shares of renewable energy (425 GW) by 2030 (in line with the Government of India's targets) will increase the likelihood that renewable generation meets marginal demand, decreasing median CO<sub>2</sub> MEFs across all states by approximately a factor of 2. Higher marginal renewable generation (mostly from solar) during the middle of the day drives this potential future decrease. Moreover, this is consistent with CO<sub>2</sub> MEFs from the United States where marginal CO<sub>2</sub> emissions decrease with increasing marginal wind generation during the middle of the day.<sup>37</sup>

A future Indian grid with more renewables will also show more variability in marginal emissions estimates, both seasonally and between hours of the day. Median absolute seasonal difference increases by up to approximately a factor of 2. Likewise, time-of-day variability increases: differences between lowest and highest median time-of-day MEFs

increases by up to a factor of approximately four. Nonetheless, under projected capacity expansions to 2030, coal generation will still form a sizable portion of marginal generation during times other than the middle of the day, especially in larger, higher demand states. This results in higher marginal emissions during these times of the day.

Comparing current and future estimates, we show stronger relationships between greater shares of average zero emission generation (mostly renewables) and lower marginal emission factors. This implies that long-term policy targets to increase renewable generation do have short-term impacts on emissions from new additional loads. Less emissions-intensive generation is more likely to meet these new loads which may not be uniformly spread across all seasons and times-of-day.

Analyzing marginal emissions as a function of current and future demand assumptions in high renewable states indicates variability and nonmonotonic relationships between demand and marginal emissions. Moreover, analyzing MEFs in this way yields estimates with wider ranges than those estimated through regression, largely due to the coincidence of marginal renewable generation and marginal demand throughout the day. However, increased marginal renewable generation yields more variability in both MEFs as a function of demand and regression-based MEFs. While all states see increases in demand from 2017–2018 to 2030–2031, no clear patterns emerge on whether marginal emissions from the future demand are distinctly lower than those from the current demand when quantifying MEFs as functions of the demand. For example, in Madhya Pradesh, 2030–2031 marginal emissions-demand curves show comparable emission factors as those for 2017–2018. On the other hand, Telangana shows decreases in dry seasons for both modeling periods. A limitation with estimating marginal emissions as a function of demand is that these curves remove the temporal correlation between hour-to-hour changes in demand at different levels by ordering the demand of different hours as a continuous series of data.

One limitation we identify is the lack of observed CEMS data to compare our 2017–2018 emissions estimates. We discuss the 2017–2018 model simulation's ability to capture temporal and spatial trends with comparisons to reported generation data in the [Supporting Information](#). Overall, the reduced-form model captures spatial and temporal trends in generation well. The weakest areas of performance are hourly generation during the monsoon season and monthly and annual generation from gas. When comparing hourly generation values to reported 2018–2019 hourly generation values, correlations between model estimates and reported values decrease due to uncertainty from renewable and hydro generations.

Our estimates for the lowest MEFs are the most uncertain with low regression  $r^2$  values and wide standard errors (see [Tables S2–S9](#)). In addition to yielding more variable and generally lower estimated MEFs, greater amounts of renewable energy that meet marginal electricity demand yield more uncertain 2030–2031 MEF estimates. This increased uncertainty furthermore highlights the limitation from the lack of CEMS data and underscores the need to have transparent, readily available empirical measurements of emissions from Indian power generation. Most analyses of Indian power generation emissions adopt bottom-up, fuel-based approaches like this analysis does.<sup>18,59</sup> In 2017–2018, large amounts of marginal coal generation yield high correlations between

marginal emissions and marginal generation to meet marginal demand, i.e., regression-based MEFs with strong fits. However, as zero-emission renewable generation increases in India for 2030–2031, we see correlations associated with MEFs degrade due to the weaker relationship between marginal generation and marginal emissions.

Previous estimates of MEFs from the United States (where empirical estimates of emissions from power plants are readily available) show that grids with higher marginal renewable generation<sup>37,39</sup> and higher marginal gas generation<sup>34</sup> yield more uncertain MEFs with lower correlations. Consequently, while regressions using empirical emissions values will not eliminate uncertainty in MEFs, they will yield more precise estimates of MEFs than those estimated off regressed bottom-up values based on assumed fuel composition and generator efficiency.

Future analyses to update MEFs for a changing Indian power system will likely need readily available CEMS data,<sup>24</sup> but another approach is emissions estimates constrained with remote sensing (satellites), with the newest generation of satellites capable of monitoring CO<sub>2</sub> from space.<sup>76–80</sup> While previous remote-sensing constrained efforts have quantified total emissions,<sup>20,21,81</sup> future areas of research will need to pair these remote-sensing methods to power plant operations to quantify the time-of-day and seasonal trends of the entire Indian grid.

Another limitation we identify are assumptions about installed capacity, demand, and market structure which increase uncertainty in our 2030–2031 estimates. We use estimates from Spencer et al. (2020)<sup>45</sup> who assume increased electricity demand due to economic growth, capacity expansion from all fuels to meet this demand, transmission capacity expansion, and further integrated power markets between states in India with the absence of current decentralized state-wise dispatch, i.e., self-scheduling. This means the future 2030–2031 India grid analyzed here has more ability than the current 2017–2018 grid to transfer power between states having heterogeneous electricity mixes. Consequently, assuming renewable capacity as Spencer et al. (2020)<sup>45</sup> do in their baseline scenario (Table S1), our 2030–2031 estimates likely characterize an upper bound in terms of the MEF decreases we see between 2017 and 2018 and 2030–2031 estimates. Greater renewable capacity would likely decrease MEFs further but increase variability and uncertainty with MEF estimates. Greater electricity sharing between states is incumbent upon policy reforms and increased transmission capacity. These estimates show an Indian grid with higher renewable generation, but several exogenous factors such as economic growth and access to capital to finance capacity and transmission expansion will likely determine the Indian electricity fuel mix in 2030. While demand is expected to grow, diurnal demand profiles may change, which would affect MEFs as well. Previous analyses and Government of India policy efforts point to air conditioning loads in the evening and charging of electric vehicles possibly changing diurnal demand profiles.<sup>82–84</sup> Moreover, the speed of current policy efforts to coordinate dispatch<sup>27</sup> between states in India will determine how the Indian power system will adopt the centralized dispatch simulated by the full-form model. If states stay fragmented in their scheduling decisions, then zero-emission electricity is less likely to move to emissions intensive states.

Along with these MEFs, we recommend users continue to use average emission factors in India at the national, regional,

and state levels to present a range of emissions outcomes from interventions.<sup>35</sup> Given limitations outlined previously, we view the MEFs generated from this analysis as a first step to quantify emissions impacts from interventions that change electricity demand. Future work will need to refine these estimates further through sensitivities scenarios. Spencer et al. (2020)<sup>45</sup> present several sensitivity scenarios with different assumptions on transmission and renewable energy capacity that can be a starting point for future work. Likewise future work will need to evaluate different diurnal demand profiles. Refining estimates is especially relevant in states and times when renewable energy is high, inducing negative or very low MEFs estimated from regressions with lower  $r^2$  values.

In conclusion, we present the first-ever spatially and temporally resolved marginal CO<sub>2</sub>, SO<sub>2</sub>, NO<sub>x</sub>, and PM<sub>2.5</sub> emission factors for current and future Indian power generation. Our results are broadly applicable to energy researchers, policy analysts, and decision makers when evaluating any interventions that change power demand in India. Given the country's growing power demand, expansion of renewable generating capacity, and state-level heterogeneity in fuel mixes and power sector operations, a detailed evaluation of emissions from new loads is critical to understanding emissions trajectories for India. Moreover, as both developed and developing countries look to net-zero targets to reduce climate-warming emissions, MEFs presented here provide once piece of information to accurately assess policies to meet these targets.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.1c07500>.

Raw CO<sub>2</sub>, NO<sub>x</sub>, SO<sub>2</sub>, and PM<sub>2.5</sub> marginal emission factors; comparisons to reported data; marginal fuel shares; input capacity for 2017–2018 and 2030–2031; diurnal demand for each state; imported and exported shares of electricity; map of India with full state names and abbreviations; and marginal generator frequency and emissions (PDF)

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## Notes

The authors declare no competing financial interest.  
Marginal emission factor data are available at [10.1184/R1/19359989](https://pubs.acs.org/10.1184/R1/19359989). All inputs, model code, and outputs for the 2017–2018 reduced form model are available at [10.1184/R1/14842224.v1](https://pubs.acs.org/10.1184/R1/14842224.v1). Output for the 2030–2031 full-form model is available at <http://103.11.86.171/home>

## ACKNOWLEDGMENTS

We thank Thomas Deetjen who provided advice to calculate our emission factors. This material is based upon work supported by the National Science Foundation Graduate Research Fellowship Program Grant No. DGE-1252522 and Grant No. DGE-1745016. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation. This publication was developed as part of the Center for Air, Climate, and Energy Solutions (CACES), which was supported under Assistance Agreement No. R835873 awarded by the U.S. Environmental Protection Agency. It has not been formally reviewed by EPA. The views expressed in this document are solely those of authors and do not necessarily reflect those of the Agency. EPA does not endorse any products or commercial services mentioned in this publication. This work was supported by the Center for Climate and Energy Decision Making (Grant SES-1463492), through a cooperative agreement between the National Science Foundation and Carnegie Mellon University. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation. Funding for this research was supported by the National Security Education Program's Boren Fellowship.

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